

Nodal Analysis (Node Voltage Method)

1. High-Level Overview & Big-Picture Intuition

What is Nodal Analysis?

Nodal Analysis (also called the **Node Voltage Method**) is a systematic technique used to analyze electrical circuits. Its main purpose is to determine the unknown **voltages** at various connection points (called **nodes**) throughout a circuit. Once you know these node voltages, you can easily calculate the current flowing through any component or the power absorbed/delivered by any element using Ohm's Law.

Fundamental Concepts Needed Before Starting

1. **Node**: Any junction or connection point where two or more circuit elements (resistors, voltage sources, current sources) meet.
2. **Reference Node (Ground / Ref)**:
 - Electric potential (voltage) is always measured *relative* to another point.
 - In Nodal Analysis, we pick **one node** as our absolute baseline reference and assign its voltage to be 0 V.
 - This baseline node is called the **Ground Node** or **Reference Node**.
3. **Node Voltage** ($V_a, V_b, V_c, \dots$): The voltage potential at a specific non-reference node measured relative to the reference node (0 V).
4. **Kirchhoff's Current Law (KCL)**:
 - KCL states that the total current entering a node must equal the total current leaving that node.
 - Mathematically:

$$\sum I_{\text{entering}} = \sum I_{\text{leaving}} \quad \text{or} \quad \sum I_{\text{leaving}} = 0$$

- *Analogy*: Think of a pipe junction carrying water. The amount of water flowing into the junction every second must exactly equal the amount of water flowing out. Water cannot accumulate inside the junction.

5. **Ohm's Law across a Branch**:
 - Current flows from higher electric potential to lower electric potential.
 - If a resistor R is connected between a high-potential node V_{high} and a low-potential node V_{low} , the current I flowing from V_{high} to V_{low} is given by:

$$I = \frac{V_{\text{high}} - V_{\text{low}}}{R}$$

2. Step-by-Step Method Breakdown

To solve any circuit using Nodal Analysis, follow these systematic steps:

Step 1: Source Transformation (Convert Voltage Sources to Current Sources)

- If the circuit contains practical voltage sources (a voltage source V_S in series with a resistor R_S), convert them into equivalent current sources:

$$I_S = \frac{V_S}{R_S}$$

- Connect the converted current source I_S in **parallel** with the resistor R_S .
- *Why do this?* Standard Nodal Analysis works smoothly with current sources because KCL relies on summing currents at nodes.

Step 2: Identify Nodes and Choose a Reference Node

- Identify all essential nodes (junctions connecting 3 or more branches).
- Select **one** node as the **Reference Node** and set its voltage to 0 V (ground: $\frac{1}{\infty}$).

- *Tip for picking reference node:* Choose the node connected to the most branches, or the node at the bottom of the circuit.
- Label all remaining non-reference nodes as $V_a, V_b, V_c, \dots$

Step 3: Write KCL Equations for Each Non-Reference Node

- For each non-reference node, assume all branch currents leave the node unless a fixed current source defines the direction.
- Express every branch current in terms of node voltages using Ohm's Law:

$$\text{Branch Current} = \frac{V_{\text{node}} - V_{\text{adjacent}}}{R}$$

- Set the sum of all leaving currents equal to zero.
- Group coefficients for $V_a, V_b, V_c, \dots$ to form clear algebraic equations.

Step 4: Solve the System of Linear Equations

- Express the equations in matrix form or solve them using algebraic elimination/substitution (or Cramer's Rule / matrix inversion) to find the unknown node voltages.

Step 5: Compute Required Quantities

- Calculate desired currents ($I = \frac{\Delta V}{R}$) or power dissipation ($P = I^2 R = \frac{V^2}{R}$).

3. General Circuit Walkthrough & Derivation

Let us demonstrate the method using a general bridge circuit containing a galvanometer/resistor network.

Step 1 Conversion (Practical Voltage Source to Current Source)

 Diagram: Source Transformation of Bridge Circuit

Visual Description:

The original diagram (left) shows a standard Wheatstone bridge configuration. Four resistors R_1, R_2, R_3, R_4 form a diamond shape with a galvanometer (internal resistance R_5 or labeled G) across the center. A voltage source V_S in series with resistor R_S is connected across the bottom nodes.

The transformed diagram (right) shows the practical voltage source (V_S in series with R_S) replaced by an equivalent current source $I_S = V_S/R_S$ placed in parallel with resistor R_S at the bottom. The current arrow points towards the left node.

Step 2 Node Labeling and Reference Selection

 Diagram: Node Labeling and Reference Assignment

Visual Description:

The circuit now has four primary nodes identified:

- **Top node:** Labeled with voltage V_b .
- **Left node:** Labeled with voltage V_a .
- **Right node:** Labeled with voltage V_c .
- **Bottom node:** Marked with the ground symbol ($\perp$) and designated as the Reference Node (0 V).
Resistor R_1 connects V_a to V_b ; R_2 connects V_a to Ground (0 V); R_3 connects V_b to V_c ; R_4 connects V_c to Ground (0 V); galvanometer resistance R_5 connects V_b to Ground (0 V); parallel resistor R_S and current source I_S connect V_a to V_c .

Step 3 Writing the KCL Node Equations

 Diagram: Applying KCL at Node A

Visual Description:

The node V_a highlight shows currents leaving toward V_b via R_1 , toward Ground (0 V) via R_2 , and toward V_c via R_S . Additionally, current I_S enters node V_a from current source I_S (directed from node V_c to V_a).

Let us formulate the equations rigorously:

KCL at Node a (V_a):

Sum of currents leaving Node $a = 0$:

Current through R_1 + Current through R_2 + Current through R_S + Current entering from source $I_S = 0$

$$\frac{V_a - V_b}{R_1} + \frac{V_a - 0}{R_2} + \frac{V_a - V_c}{R_S} + I_S = 0$$

Rearranging terms by grouping V_a , V_b , and V_c :

$$\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_S} \right) V_a - \left(\frac{1}{R_1} \right) V_b - \left(\frac{1}{R_S} \right) V_c = -I_S$$

KCL at Node b (V_b):

Sum of currents leaving Node $b = 0$:

$$\frac{V_b - V_a}{R_1} + \frac{V_b - 0}{R_5} + \frac{V_b - V_c}{R_3} = 0$$

Rearranging terms:

$$- \left(\frac{1}{R_1} \right) V_a + \left(\frac{1}{R_1} + \frac{1}{R_5} + \frac{1}{R_3} \right) V_b - \left(\frac{1}{R_3} \right) V_c = 0$$

KCL at Node c (V_c):

Sum of currents leaving Node $c = 0$:

$$\frac{V_c - V_a}{R_S} + \frac{V_c - V_b}{R_3} + \frac{V_c - 0}{R_4} - I_S = 0$$

Rearranging terms:

$$- \left(\frac{1}{R_S} \right) V_a - \left(\frac{1}{R_3} \right) V_b + \left(\frac{1}{R_S} + \frac{1}{R_3} + \frac{1}{R_4} \right) V_c = I_S$$

4. Fully Worked Examples

Illustration 1: Galvanometer Current in a Bridge Circuit

 Diagram: Illustration 1 Bridge Circuit

Visual Description:

A Wheatstone bridge circuit powered at the bottom by a practical voltage source (10 V source in series with 10 Ω resistor).

- Top node is labeled **A**.
- Bottom interior node is labeled **B**.
- A galvanometer with resistance $R_G = 100 \Omega$ connects node A and node B vertically.
- Left upper branch: 40 Ω .

- Right upper branch: $30\ \Omega$.
- Left lower branch: $20\ \Omega$.
- Right lower branch: $30\ \Omega$.
- Goal: Find current I_{AB} through the galvanometer from Node A to Node B.

Step-by-Step Solution:

Step 1: Convert Practical Voltage Source to Current Source

- Voltage source $V_S = 10\text{ V}$ in series with $R_S = 10\ \Omega$.
- Equivalent current source:

$$I_S = \frac{V_S}{R_S} = \frac{10\text{ V}}{10\ \Omega} = 1\text{ A}$$

- Resistor $R_S = 10\ \Omega$ is now in parallel with current source $I_S = 1\text{ A}$, connected between the left node (C) and right node (D).

Step 2: Assign Nodes and Reference

- Node A = V_A (Top node)
- Node B = Reference Node = 0 V (Bottom junction inside bridge)
- Node C = V_C (Leftmost junction)
- Node D = V_D (Rightmost junction)

Note: Alternatively, choose Node B as reference 0 V .

Let's designate Node B as Ground ($V_B = 0\text{ V}$).

Remaining unknown node voltages: V_A , V_C , V_D .

Step 3: Write KCL Equations

1. At Node A (V_A):

$$\frac{V_A - V_C}{40} + \frac{V_A - 0}{100} + \frac{V_A - V_D}{30} = 0$$

Multiply entire equation by 600 (LCM) to clear denominators:

$$15(V_A - V_C) + 6V_A + 20(V_A - V_D) = 0$$

$$15V_A - 15V_C + 6V_A + 20V_A - 20V_D = 0$$

$$41V_A - 15V_C - 20V_D = 0 \quad \text{--- (Equation 1)}$$

2. At Node C (V_C):

$$\frac{V_C - V_A}{40} + \frac{V_C - 0}{20} + \frac{V_C - V_D}{10} + (-1) = 0$$

Multiply entire equation by 40:

$$(V_C - V_A) + 2V_C + 4(V_C - V_D) - 40 = 0$$

$$-V_A + 7V_C - 4V_D = 40 \quad \text{--- (Equation 2)}$$

3. At Node D (V_D):

$$\frac{V_D - V_A}{30} + \frac{V_D - 0}{30} + \frac{V_D - V_C}{10} + 1 = 0$$

Multiply entire equation by 30:

$$(V_D - V_A) + V_D + 3(V_D - V_C) + 30 = 0$$

$$-V_A - 3V_C + 5V_D = -30 \quad \text{--- (Equation 3)}$$

Step 4: Solve System of Simultaneous Equations

From Equation 2:

$$7V_C = 40 + V_A + 4V_D \implies V_C = \frac{40 + V_A + 4V_D}{7}$$

Substitute into Equation 3:

$$-V_A - 3\left(\frac{40 + V_A + 4V_D}{7}\right) + 5V_D = -30$$

Multiply by 7:

$$-7V_A - 120 - 3V_A - 12V_D + 35V_D = -210$$

$$-10V_A + 23V_D = -90 \implies V_D = \frac{10V_A - 90}{23}$$

Substitute V_C and V_D back into Equation 1 and solve for V_A :

$$V_A = 0.98 \text{ V}$$

Step 5: Calculate Galvanometer Current I_{AB}

Current through galvanometer from Node A to Node B ($V_B = 0 \text{ V}$):

$$I_{AB} = \frac{V_A - V_B}{R_G} = \frac{0.98 \text{ V} - 0 \text{ V}}{100 \Omega} = 0.0098 \text{ A} = 9.8 \text{ mA}$$

Illustration 2: Power Dissipated in a Resistor

 Diagram: Illustration 2 Circuit Layout

Visual Description:

Circuit layout containing:

- A DC voltage source 10 V on the lower left connected to ground.
- A top branch containing an 8 V source in series with a 2Ω resistor.
- A central horizontal branch with a 4Ω resistor.
- An independent current source of 2 A directed upwards.
- Two parallel vertical resistors on the right: 6Ω and 8Ω .
- Bottom wire is connected continuously, acting as ground.

Step-by-Step Solution:

Step 1: Identify Nodes and Assign Reference

- **Bottom common line:** Set as Reference Node (0 V).
- **Node 1 (Left Node, above 10V source):**

Since a 10 V ideal voltage source is connected directly between Node 1 and Ground (0 V):

$$V_1 = 10 \text{ V}$$

- **Node 2 (Middle-Right Node):**

Notice that the wire above 4 Ω , 2 A source, 6 Ω , and 8 Ω connects into a single major junction node. Let's call its voltage V_2 .

Step 2: Write KCL Equation at Node 2 (V_2)

All currents leaving Node 2 must sum to zero:

Current through top branch + Current through 4 Ω + Current into 2A source + Current through 6 Ω + Current through 8 Ω = 0

1. **Top Branch (connected between $V_1 = 10 \text{ V}$ and V_2 via 8 V source and 2 Ω resistor):**

$$\text{Current leaving } V_2 = \frac{V_2 - (V_1 - 8)}{2} = \frac{V_2 - (10 - 8)}{2} = \frac{V_2 - 2}{2}$$

2. **Branch through 4 Ω resistor (connected to $V_1 = 10 \text{ V}$):**

$$\text{Current leaving } V_2 = \frac{V_2 - V_1}{4} = \frac{V_2 - 10}{4}$$

3. **Branch through 2 A current source:**

Since 2 A enters Node 2, leaving current is -2 A .

4. **Branch through 6 Ω resistor (connected to Ground 0 V):**

$$\text{Current leaving } V_2 = \frac{V_2 - 0}{6} = \frac{V_2}{6}$$

5. **Branch through 8 Ω resistor (connected to Ground 0 V):**

$$\text{Current leaving } V_2 = \frac{V_2 - 0}{8} = \frac{V_2}{8}$$

Combine all branch equations into KCL:

$$\frac{V_2 - 2}{2} + \frac{V_2 - 10}{4} - 2 + \frac{V_2}{6} + \frac{V_2}{8} = 0$$

Step 3: Solve for V_2

Find a common denominator for (2, 4, 6, 8), which is 24. Multiply entire equation by 24:

$$24 \cdot \left(\frac{V_2 - 2}{2} \right) + 24 \cdot \left(\frac{V_2 - 10}{4} \right) - 24 \cdot (2) + 24 \cdot \left(\frac{V_2}{6} \right) + 24 \cdot \left(\frac{V_2}{8} \right) = 0$$

$$12(V_2 - 2) + 6(V_2 - 10) - 48 + 4V_2 + 3V_2 = 0$$

$$12V_2 - 24 + 6V_2 - 60 - 48 + 4V_2 + 3V_2 = 0$$

$$(12 + 6 + 4 + 3)V_2 - (24 + 60 + 48) = 0$$

$$25V_2 - 132 = 0$$

$$25V_2 = 132 \implies V_2 = \frac{132}{25} = 5.28 \text{ V}$$

Step 4: Calculate Power Dissipated in 8 Ω Resistor

The voltage across the 8 Ω resistor is simply $V_2 - 0 = 5.28 \text{ V}$.

Using the power formula $P = \frac{V^2}{R}$:

$$P_{8\Omega} = \frac{(V_2)^2}{8} = \frac{(5.28)^2}{8} = \frac{27.8784}{8} = 3.4848 \text{ W} \approx 3.48 \text{ W}$$

5. Summary Cheat-Sheet for Exam Preparation

Step	Action	Formulas / Key Rules
1	Convert Voltage Sources	$I_S = \frac{V_S}{R_S}$ in parallel with R_S
2	Pick Reference Node	Set $V_{\text{ref}} = 0 \text{ V}$ (Ground)
3	Apply KCL at Nodes	$\sum I_{\text{leaving}} = 0 \implies \sum \frac{V_{\text{node}} - V_{\text{adjacent}}}{R} = 0$
4	Express in Matrix Form	$[G][V] = [I]$ where G is conductance matrix
5	Calculate Required Variables	Power $P = \frac{V^2}{R} = I^2 R$; Branch Current $I = \frac{\Delta V}{R}$